

“... the invention of paramount value, appears to me to be this cartridge...”

CAPTAIN O'HEA, *THE JOURNAL OF THE SOCIETY OF ARTS* (1867)

Unitary metallic cartridges transformed conflicts in the late 19th century. They played a key role in the Battle of Hoover's Gap—a decisive engagement of the Tullahoma Campaign in the American Civil War (1861–65). The Union Army was outnumbered by Confederate forces, which were in a strong defensive position. Marching rapidly into Hoover's Gap, the Union forces surprised the Confederates who scattered initially. In the battle that ensued, the Confederate Army regrouped and charged at the Union soldiers. Despite facing a volley of gunfire, the Confederate soldiers continued to advance, not expecting the Union rifles to be reloaded quickly. However, the Union soldiers were armed with new Spencer repeating rifles loaded

with .56in-caliber rim-fire cartridges. These weapons could fire more than 14 rounds per minute and proceeded to cut down almost one-quarter of the Confederate Army.

In the Anglo-Zulu War (1879), a small number of British soldiers used the new technology in a similar way. Armed with Martini-Henry rifles loaded with Boxer cartridges, they repelled a vast Zulu army against all odds, because they were able to reload and fire swiftly in the heat of battle. Armed with superior guns and ammunition, European powers scrambled to make forays into Africa at the turn of the 20th century.

These conflicts exemplified the advantages of the metallic cartridge, without which self-loading and automatic firearms would not have seen the light of day.

KEY FIGURE

Hiram Berdan
(1824–93)

Engineer and inventor Hiram Berdan was a colonel of the United States Volunteer Sharpshooter Regiments during the American Civil War. A sought-after weapons designer, he was commissioned by the Russian Army to update its infantry firearms. He created the Berdan cartridge, which would go on to become the standard for metallic cartridges seen today.



AFTER »

Once the idea of self-contained ammunition had taken hold, cartridges and their associated firearms underwent a long process of evolution, eventually resulting in the invention of repeating rifles (see p. 116) and magazine feeding systems.

• **EARLY CENTER-FIRE CARTRIDGES**, such as the .450in Martini Henry Boxer cartridge, were composite assemblies. The flimsy bodies were easily distorted and forcible extraction, in the heat of battle, could pull off the disk forming the rim. These problems were overcome as the composite assembly cartridges were replaced by solid-drawn cartridges.

• **MUZZLE-LOADERS WERE CONVERTED** into breech-loading weapons to utilize metallic cartridges. This spurred the refinement of breech-loading systems, resulting in the growth of more efficient breech-loading weapons, and eventually, self-loading firearms.

• **THE DURABILITY** of solid-drawn metallic cartridges allowed them to be loaded from magazines on guns. Repeating weapons designed to accept cartridges in magazines developed rapidly, leading to the firearms of today.



**.450IN MARTINI
HENRY BOXER
CARTRIDGE**

◀ DEFENDING RORKE'S DRIFT

In the defense of Rorke's Drift (1879) in the Anglo-Zulu War, fewer than 150 British soldiers defended themselves against an overwhelming force of 4,000 Zulu warriors. The use of Martini rifles and coiled brass-cased cartridges enabled the British forces to load and fire quickly, saving them from almost certain slaughter. Some soldiers can be seen handling the cartridges in the picture.

